

TrES-3b

R-2522

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110510 · created 2026-08-25T01:20:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455691.9454 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.019
Transit depth (Rp/Rs)^2	2.51 +/- 0.59 [%]
Orbital Inclination (inc)	82.54 +/- 0.62
Airmass coefficient 1 (a1)	1.109 +/- 0.011
Airmass coefficient 2 (a2)	-0.102 +/- 0.074
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	None
Optimal Aperture	3.79
Optimal Annulus	9.74
Transit Duration (day)	0.0601 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.019 vs 0.16309 (deviation 0.27 σ)
Duration fit vs published	0.0601 d vs 0.0567 d (deviation 0.62 σ)
Mid-time O–C	1.4 min
Depth SNR	8.3
Residual scatter	0.97 %

Light curve

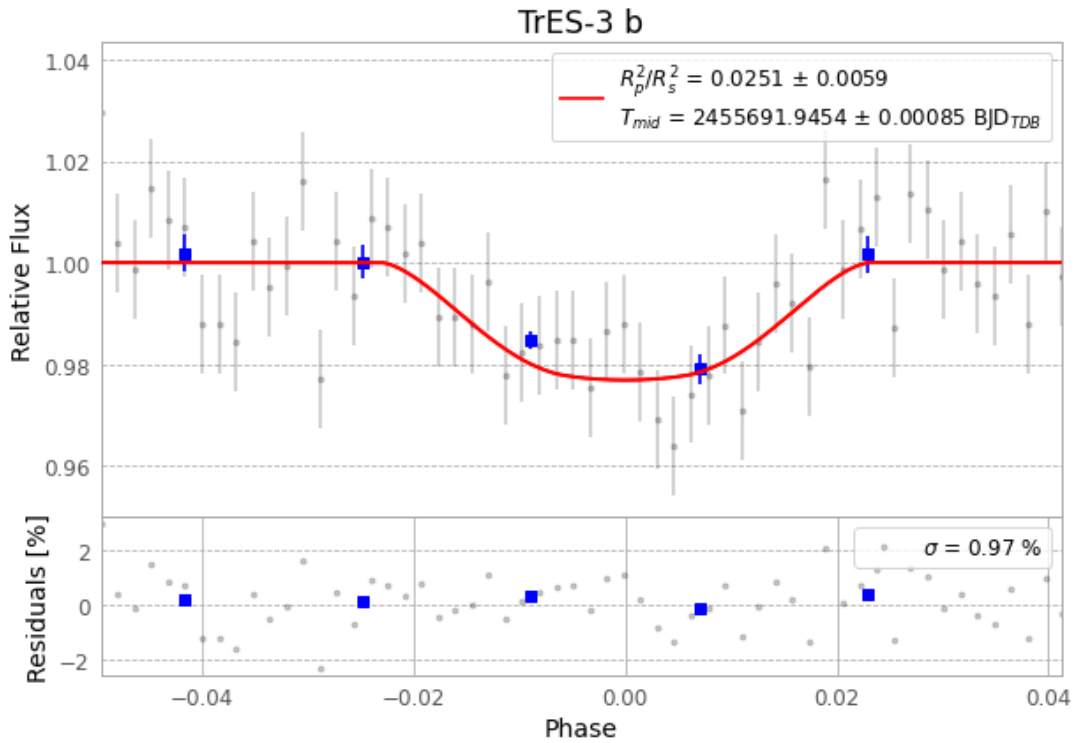


Figure 1 — FinalLightCurve_TrES-3 b_2011-05-10.png (SHA-256 6029f18e6cff099c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 194], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a51492ca32798c5d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), TRES-3110510090316.FITS → TRES-3110510120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110510.log (15f1baa50b62...), FinalLightCurve_TrES-3 b_2011-05-10.png (6029f18e6cff...), FinalLightCurve_TrES-3 b_2011-05-10.csv (e0f1ed86704d...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:20Z.